



OMNIX QUANTUM LTD
DECISION GOVERNANCE INFRASTRUCTURE

UNITED STATES PATENT AND TRADEMARK OFFICE
PROVISIONAL PATENT APPLICATION

TITLE OF INVENTION:

**SESSION-LEVEL CONTEXT ADMISSION GATE FOR
AUTOMATED DECISION PIPELINES WITH
FOUR-DIMENSIONAL GLOBAL CONDITION EVALUATION,
MACRO-RISK CIRCUIT BREAKER, AND EPISTEMIC
TRANSPARENCY ENFORCEMENT**

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PROVISIONAL PATENT APPLICATION

OMNIX-PAT-2026-006

Title: SESSION-LEVEL CONTEXT ADMISSION GATE FOR AUTOMATED DECISION PIPELINES WITH FOUR-DIMENSIONAL GLOBAL CONDITION EVALUATION, MACRO-RISK CIRCUIT BREAKER, AND EPISTEMIC TRANSPARENCY ENFORCEMENT

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FIELD OF THE INVENTION

The present invention relates to automated decision governance systems, and more particularly to a session-level pre-admission gate that evaluates global environmental conditions before any individual decision is evaluated by a governance pipeline, preventing initiation of decision evaluation sessions when macro-level conditions render the environment structurally inadmissible for automated decision-making.

BACKGROUND

I. THE GAP BETWEEN PER-DECISION GOVERNANCE AND MACRO ADMISSIBILITY

Existing automated decision governance systems evaluate the quality, risk, and admissibility of individual proposed decisions. These systems assess whether a specific proposed action — given the current signal values, risk exposure, and coherence of the signals supporting that action — meets the requirements for governance certification.

However, this per-decision evaluation approach contains a structural gap: it evaluates whether a specific decision is admissible given current conditions, without first evaluating whether current conditions make any decision admissible at all.

Consider an automated trading system operating during a market-wide liquidity crisis. Individual proposed trades may appear to satisfy all per-decision governance criteria — the probability score is above threshold, the risk exposure is within configured limits, the signals are coherent. Yet all of these certifications are rendered meaningless by the fact that the global market environment has become so extreme that automated decision-making is structurally inappropriate regardless of the apparent quality of any individual decision.

This problem is not limited to financial markets. In clinical decision support systems, a system operating during a data infrastructure failure may produce individual recommendations that appear technically valid while the underlying data sources are corrupted. In autonomous robotics, a system operating in an environment that has changed structurally (e.g., unexpected obstacles, changed physical layout) may produce individually valid movement commands that are collectively dangerous in the changed environment.

II. INADEQUACY OF EXISTING APPROACHES

2.1 Per-Decision-Only Governance: All known prior art governance systems evaluate governance at the level of individual decisions. No known prior art system implements a session-level admission gate that must be passed before any

per-decision evaluation begins.

2.2 Implicit Macro-Level Assumptions: Existing systems implicitly assume that macro-level conditions are suitable for automated decision-making at all times. This assumption is embedded in the system architecture — by proceeding directly to per-decision evaluation, the system treats the absence of explicit macro-level checking as implicit approval of the macro environment.

2.3 False Score Fabrication: Some existing systems assign high "admission scores" or "confidence scores" when a gate module is disabled, treating the absence of gate evaluation as evidence of favorable conditions. This conflates "not evaluated" with "passed evaluation" — a logical error that fabricates system confidence without evidentiary basis.

2.4 No Regulatory Alignment for Macro Circuit Breakers: Existing automated decision systems do not implement the circuit breaker obligations required by financial regulations (MiFID II) or the context-aware risk management required by AI governance frameworks (NIST AI RMF, EU AI Act Article 9) at the session-initiation level.

The present invention addresses all four inadequacies through the Context Admission Gate (CAG) architecture.

SUMMARY OF THE INVENTION

The present invention provides a session-level context admission gate that evaluates global environmental conditions before any individual decision enters an automated governance pipeline. The gate:

- Evaluates four independent global condition dimensions: global volatility, cross-pair correlation, liquidity score, and macro risk, each against an independently configurable threshold.
- Issues a session-level admission or block decision that applies to the entire evaluation session — not to individual decisions within the session.
- Enforces epistemic transparency by explicitly distinguishing between session-admitted, session-blocked, disabled (not evaluated), and failsafe (error) states, preventing false confidence fabrication.
- Operates as an independent pre-pipeline layer — architecturally upstream of all per-decision checkpoints — ensuring that per-decision evaluation cannot proceed when global conditions are inadmissible.
- Produces structured admission records suitable for inclusion in post-quantum cryptographic governance receipts.

DETAILED DESCRIPTION OF THE PREFERRED EMBODIMENTS

I. ARCHITECTURAL POSITION AND DISTINCTION FROM PER-DECISION GOVERNANCE

The Context Admission Gate (hereinafter "CAG" or "the System") occupies a specific architectural position immediately upstream of the per-decision governance checkpoint pipeline:

AVM → CAG → Sequential Checkpoint Pipeline → Individual Decision

This position is architecturally definitive. The CAG evaluates whether the current global environment permits the initiation of any decision evaluation session. The per-decision checkpoint pipeline — which evaluates whether a specific proposed decision is admissible — operates only after the CAG has admitted the session.

The Critical Architectural Distinction: The per-decision checkpoint pipeline evaluates WHAT to decide: it assesses signal quality, probability scores, risk exposure, coherence, and decision consistency for a specific proposed action. The CAG evaluates WHETHER the global context allows any decision at all: it assesses whether macro-level environmental conditions are suitable for automated decision-making regardless of what the specific proposed decision is.

This distinction is absolute. A proposed decision that passes all per-decision governance checks is inadmissible if the CAG has blocked the session. Conversely, a session admitted by the CAG proceeds to per-decision evaluation — CAG admission does not certify individual decisions; it permits the evaluation process to begin.

The System is applicable to any domain in which automated decisions are made in an environment that may become macro-level inadmissible, including but not limited to: financial trading during market crises or regulatory halts; clinical

decision support during data infrastructure failures; autonomous robotic systems during environmental structural changes; energy grid management during grid instability events; and autonomous agent orchestration platforms during dependency or coordination failures.

II. FOUR-DIMENSIONAL GLOBAL CONDITION EVALUATION

The CAG evaluates global conditions along four independent dimensions, each with an independently configurable threshold:

II.A. Global Volatility (Dimension 1)

Definition: A normalized measure of cross-asset or market-wide volatility, analogous to the VIX index in financial contexts but generalized to any domain where an overall system volatility metric is available.

Admission condition: `global_volatility < global_volatility_threshold`.

In one embodiment, the default threshold is 80.0 on a 0–100 normalized scale. In other embodiments, this threshold may be set to any value appropriate to the deployment domain and risk tolerance of the deploying institution. For non-financial deployments, the global volatility metric may represent: system-wide error rates in a software deployment context; patient population acuity index in a clinical context; grid frequency deviation in an energy management context; or environmental variability index in a robotics context.

Violation behavior: If `global_volatility ≥ threshold`, the System issues a session BLOCK with violation reason VOLATILITY_EXCEEDED. No further dimensions are evaluated (in one embodiment with `block_on_any_violation=True`).

II.B. Cross-Pair Correlation (Dimension 2)

Definition: A normalized measure of abnormal correlation between independent decision inputs that would ordinarily be expected to be partially uncorrelated. High cross-pair correlation indicates regime instability — a condition in which independent signals are moving together in an atypical manner, reducing the effective diversity of the information base.

Admission condition: `cross_pair_correlation < cross_pair_correlation_threshold`.

In one embodiment, the default threshold is 90.0. In other embodiments, this threshold may be calibrated for any domain in which independent input sources are expected to maintain partial decorrelation. In clinical contexts, this dimension may represent abnormal correlation among independent diagnostic indicators; in robotics, abnormal correlation among independent sensor streams.

II.C. Liquidity Score (Dimension 3)

Definition: A normalized measure of the system's operational liquidity — the availability of resources, counterparties, data sources, or execution capacity required for decisions to be implementable. Low liquidity indicates that decisions, even if individually admissible, may not be executable in the current environment.

Admission condition: `liquidity_score ≥ liquidity_score_minimum`.

Note the reversed condition: unlike dimensions 1 and 2, which must be BELOW their thresholds, the liquidity score must be ABOVE its minimum. In one embodiment, the default minimum is 20.0. In other embodiments, this minimum may be calibrated for any domain. In software deployment contexts, `liquidity_score` may represent available compute capacity; in supply chain contexts, available inventory or fulfillment capacity.

II.D. Macro Risk (Dimension 4)

Definition: A composite measure of systemic or macro-level risk in the current operating environment, derived from multiple sources and capturing risks that are not captured by any individual per-decision signal.

Admission condition: `macro_risk < macro_risk_ceiling`.

In one embodiment, the default ceiling is 85.0. The `macro_risk` metric is intended to capture risks that are systemic in nature — risks that affect the entire decision-making environment rather than individual decisions. In financial contexts, this may include regulatory halt indicators, systemic contagion signals, or cross-market stress indicators. In clinical contexts, this may include public health emergency indicators or institutional capacity crisis signals.

III. SESSION-LEVEL OPERATION AND ADMISSION SEMANTICS

III.A. Session-Level vs. Decision-Level Operation

The CAG produces one admission decision per evaluation session, not one decision per individual proposed action within the session. An evaluation session is defined as a bounded time window during which a set of related decision evaluations may be initiated. The admission decision made at session initiation applies to all individual decision evaluations within that session.

This session-level operation is architecturally distinct from all per-decision checkpoint evaluations. It reflects the principle that global environmental conditions change at a slower timescale than individual decision signals, and that the administrative overhead of re-evaluating global conditions for every individual decision is both unnecessary and operationally counterproductive.

In one embodiment, the session window duration is configurable, with a default of one evaluation period (e.g., one trading session, one shift, one operational cycle) appropriate to the deployment domain.

III.B. Block-on-Any-Violation vs. Aggregate Scoring

In one embodiment, the CAG applies a `block_on_any_violation=True` policy, meaning that any single dimension exceeding its threshold causes an immediate session BLOCK, regardless of the values of other dimensions. This is the strictest admission policy and is the default.

In other embodiments, the CAG may be configured to compute an aggregate admission score across all four dimensions and block only when the aggregate score exceeds a combined threshold. This aggregate mode may be appropriate in deployments where individual dimension violations are expected to be occasional and where the overall environmental health is more reliably captured by a combined score.

III.C. Admission Score and Structured Output

The CAG produces a structured CAGResult containing:

- **admitted:** Boolean indicating session admission or block
- **pass_through:** Boolean indicating the gate was disabled or encountered an error (not evaluated)
- **reason:** Human-readable reason for the admission decision
- **admission_score:** A numeric score reflecting the overall environmental health assessment (0–100)
- **violation:** Specific dimension and condition that caused a block, if applicable
- **gate_checks:** List of all dimension evaluations performed, for audit purposes
- **evaluation_state:** One of EVALUATED, DISABLED, or FAILSAFE

The structured output is designed for direct inclusion in governance receipts, enabling the CAG's session admission decision to be documented in the same audit trail as per-decision governance certifications.

IV. EPISTEMIC TRANSPARENCY ENFORCEMENT

A defining design principle of the System is the enforcement of epistemic transparency — the principle that the system's confidence scores and state indicators must accurately reflect the evidentiary basis for that confidence.

IV.A. Disabled State Transparency

When the CAG is disabled (`CAG_ENABLED=False` or equivalent configuration), the System returns `admission_score=0.0` and `evaluation_state="DISABLED"`.

Critical principle: A disabled gate has NOT evaluated the global environment. It would be epistemically incorrect to return a high admission score (e.g., 100.0) when the gate is disabled, as this would fabricate evidence of favorable global conditions without any actual evaluation. `admission_score=0.0` explicitly signals the absence of evaluation, not the presence of favorable conditions.

IV.B. Failsafe State Transparency

When the CAG encounters an internal error during evaluation, the System returns `admission_score=0.0` and `evaluation_state="FAILSAFE"`, and defaults to session admission (pass-through) to avoid blocking the pipeline due to module malfunction. However, the `admission_score=0.0` signals that the admission is not based on a favorable evaluation but on a failsafe default.

Architectural rationale: The failsafe-to-admit policy for the CAG differs from the fail-closed policy of the AVM. This reflects the different risk profiles: the CAG operates at a coarser level than the AVM, and a CAG module failure should not by itself halt an entire decision pipeline. The per-decision checkpoint pipeline continues to provide governance even when the CAG has failed — but the failsafe state is clearly documented.

IV.C. Evaluated State

When the CAG has performed actual evaluation of global conditions, the System returns `evaluation_state="EVALUATED"` and an `admission_score` reflecting the actual assessment. This is the only state in which the `admission_score` has evidentiary meaning.

V. REGULATORY ALIGNMENT

The CAG architecture directly implements session-level requirements from three regulatory frameworks:

NIST AI RMF: The AI Risk Management Framework requires context-aware governance — the recognition that AI system risks vary with operational context. The CAG implements context-aware governance at the session level by evaluating whether the current global context supports safe automated decision-making.

EU AI Act Article 9: Article 9 mandates risk management systems for high-impact AI. The CAG's macro-risk circuit breaker directly implements the circuit breaker obligation — automatically suspending operation when macro-level conditions exceed safety thresholds.

MiFID II Circuit Breaker Obligations: MiFID II requires automated trading systems to implement circuit breakers that halt operation during extreme market conditions. The CAG implements this obligation at the session level through the `global_volatility` and `macro_risk` dimensions.

In other embodiments, the CAG may be configured to align with domain-specific regulatory requirements, including FDA guidance on AI/ML-based software as a medical device (SaMD) for clinical deployments, or IEC 61508 functional safety standards for industrial and robotics deployments.

VI. ALTERNATIVE EMBODIMENTS

6.1 Dynamic Threshold Adjustment: In one embodiment, CAG thresholds are adjusted dynamically based on historical admission patterns, tightening thresholds during periods of elevated global risk and relaxing them during stable periods.

6.2 Multi-Dimensional Aggregate Scoring: In one embodiment, the four dimensions are combined into a weighted aggregate admission score, with weights configurable to reflect the relative importance of each dimension in the deployment domain.

6.3 Temporal Smoothing: In one embodiment, each dimension value is smoothed over a configurable temporal window (e.g., a 15-minute rolling average) before threshold comparison, preventing brief spikes from triggering unnecessary session blocks.

6.4 Domain-Specific Dimension Sets: In non-financial embodiments, the four dimensions may be replaced with domain-appropriate global condition metrics. The architectural requirement — that a session-level gate evaluates macro-level conditions before any per-decision evaluation begins — is preserved regardless of the specific metrics used.

6.5 Cascading Session Blocks: In one embodiment, a session block issued by the CAG is propagated to dependent systems, preventing downstream automated pipelines from initiating new evaluation sessions until the CAG admits a new session.

CLAIMS

Claim 1 (Broad — Core CAG) — A computer-implemented session-level admission gate for an automated decision pipeline, comprising: a global condition evaluation module that evaluates a plurality of global environmental condition metrics; an admission decision module that issues a session admission or session block decision based on said evaluation; and a pipeline interface that prevents initiation of per-decision evaluation when a session block decision has been issued; wherein said session-level admission decision applies to all individual decision evaluations within the session.

Claim 2 (Specific — Four Dimensions) — The system of Claim 1, wherein the global condition evaluation module evaluates at least: a global volatility metric against a configurable upper threshold; a cross-entity correlation metric against a configurable upper threshold; a liquidity availability metric against a configurable lower minimum; and a composite macro risk metric against a configurable upper ceiling; wherein violation of any configured threshold condition causes a session block.

Claim 3 (Broad — Architectural Distinction) — The system of Claim 1, wherein the session-level admission gate operates as a distinct architectural layer upstream of all per-decision governance checkpoints, and wherein per-decision governance evaluation is not initiated for any proposed action within a blocked session regardless of the apparent quality of individual decision signals.

Claim 4 (Broad — Epistemic Transparency) — The system of Claim 1, further comprising an epistemic transparency module that enforces distinct state representations for: a first state in which global conditions were evaluated and found admissible; a second state in which global conditions were evaluated and found inadmissible; a third state in which the gate was disabled and no evaluation was performed; and a fourth state in which the gate encountered an internal error; wherein the third and fourth states produce an admission score of zero rather than an admission score indicating favorable conditions.

Claim 5 (Specific — Disabled Score) — The system of Claim 4, wherein the third state (gate disabled) produces `admission_score=0.0` and `evaluation_state="DISABLED"`, explicitly indicating the absence of evaluation rather than fabricating evidence of favorable global conditions.

Claim 6 (Specific — Structured Output) — The system of Claim 1, wherein the admission decision module produces a structured admission record comprising: an admission boolean, a pass-through indicator, a human-readable reason, a numeric admission score, a violation description, a list of dimension evaluations performed, and an evaluation state identifier; wherein said structured record is suitable for inclusion in a cryptographic governance receipt.

Claim 7 (Broad — Multi-Domain) — The system of Claim 1, wherein the global condition metrics are configurable for deployment in any domain in which automated decisions are made in environments that may become macro-level inadmissible, including financial trading, clinical decision support, autonomous robotic systems, energy grid management, and autonomous agent orchestration platforms.

Claim 8 (Broad — Regulatory Circuit Breaker) — A computer-implemented macro-level circuit breaker for automated decision systems, comprising: a session-initiation evaluation module that assesses global environmental conditions before any automated decision evaluation begins; a blocking module that suspends all automated decision evaluation for a session when any global condition metric exceeds a configured admissibility threshold; and an audit documentation module that records the specific metric and threshold values that caused suspension; wherein the circuit breaker operates at the session level rather than the individual decision level.

Claim 9 (Broad — Method Claim) — A computer-implemented method for session-level admission control in an automated decision pipeline, comprising: receiving a request to initiate a decision evaluation session; evaluating a plurality of global environmental condition metrics independently against configurable thresholds; issuing a session block when any metric violates its threshold; permitting per-decision evaluation to proceed only for admitted sessions; and recording a structured admission decision with explicit evaluation state for inclusion in governance audit records.

Claim 10 (Broad — System Claim) — A computer-implemented system configured to enforce macro-level admissibility as a precondition for automated decision-making by: evaluating global environmental conditions at session initiation independently of any per-decision evaluation; blocking entire decision evaluation sessions when global conditions are

macro-inadmissible; distinguishing explicitly between evaluated-admitted, evaluated-blocked, not-evaluated-disabled, and not-evaluated-failsafe states; and producing structured session admission records with zero admission scores for non-evaluated states, preventing fabrication of environmental confidence without evidentiary basis.

ABSTRACT

A session-level context admission gate enforces macro-level admissibility as a precondition for automated decision-making across any domain. The gate evaluates four independent global condition dimensions — global volatility, cross-entity correlation, liquidity availability, and composite macro risk — against independently configurable thresholds at session initiation, before any individual decision enters the per-decision governance pipeline. A session block suspends all decision evaluation for the session regardless of individual decision quality. The gate enforces epistemic transparency by assigning zero admission scores when disabled or in failsafe states, preventing fabrication of environmental confidence without evaluation evidence. Structured admission records document the evaluation state, specific violation metrics, and dimension-by-dimension check results for inclusion in cryptographic governance receipts. The architecture is applicable to financial trading, clinical decision support, autonomous robotics, energy grid management, and autonomous agent orchestration, and implements session-level circuit breaker obligations required by MiFID II, NIST AI RMF, and EU AI Act Article 9. To the inventor's knowledge, this is the first system to implement a session-level pre-admission gate as a distinct architectural layer upstream of per-decision governance checkpoints, with explicit epistemic transparency enforcement distinguishing evaluated from non-evaluated admission states.

DRAWINGS

FIG. 1 – CAG Architectural Position

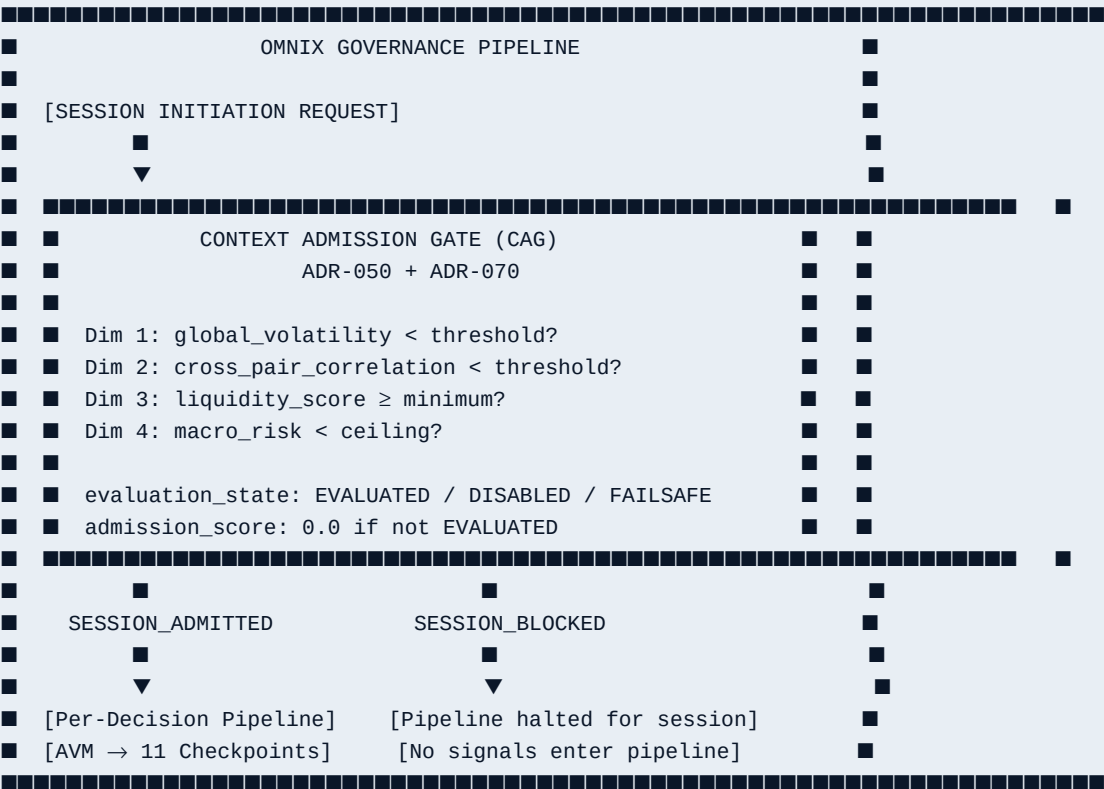


FIG. 2 – Epistemic Transparency State Matrix

State	admitted	pass_through	score	evaluation_state
ADMITTED	True	False	>0	EVALUATED
BLOCKED	False	False	0	EVALUATED
DISABLED	True	True	0	DISABLED
FAILSAFE	True	True	0	FAILSAFE

Rule: score > 0 IFF evaluation_state == EVALUATED AND admitted

FIG. 3 – Four-Dimension Evaluation (Default Thresholds – One Embodiment)

Dimension	Condition	Default Threshold
global_volatility	< threshold	80.0
cross_pair_correlation	< threshold	90.0
liquidity_score	≥ minimum	20.0
macro_risk	< ceiling	85.0

All configurable. Domain-specific calibration required.
block_on_any_violation=True (default): first failure → SESSION_BLOCKED